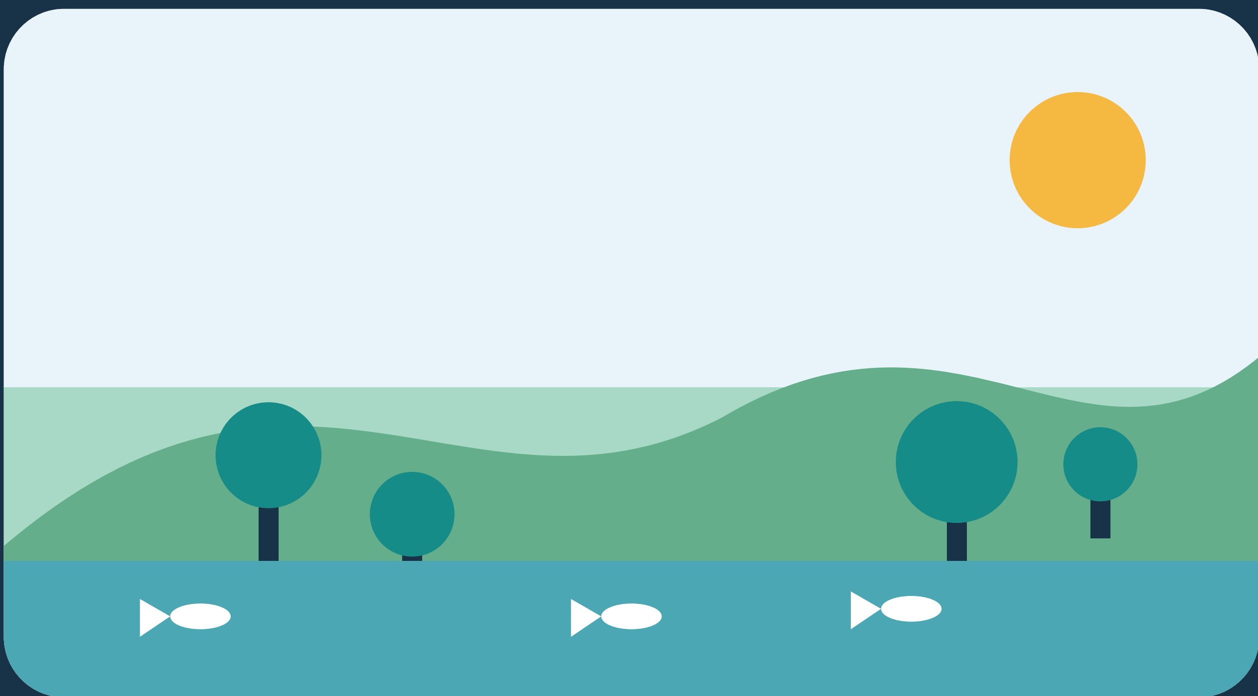
FOUNDATIONS OF ENVIRONMENTAL SCIENCE

Introduction to Ecosystems

Energy, cycles, and the living world



STUDENT EDITION | SAMPLE TEXTBOOK FIXTURE

What Is an Ecosystem?

An ecosystem is a community of organisms interacting with one another and with the nonliving world around them.

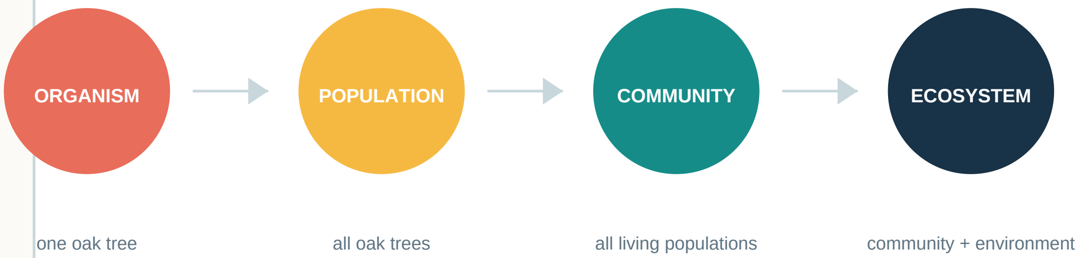
KEY VOCABULARY

biotic - living or once-living parts **abiotic** - nonliving physical factors
population - members of one species **community** - interacting populations

A woodland, a tide pool, and even a small garden can be studied as ecosystems. Their boundaries are chosen by the observer, but the relationships inside them are real: organisms need space, materials, and energy, while temperature, water, light, and soil shape which organisms can thrive.

Scientists often begin by sorting observations into two categories. The biotic category includes plants, animals, fungi, and microbes. The abiotic category includes sunlight, air, minerals, water, and climate. Neither category works alone; an ecosystem emerges from their continuous interaction.

Levels of ecological organization

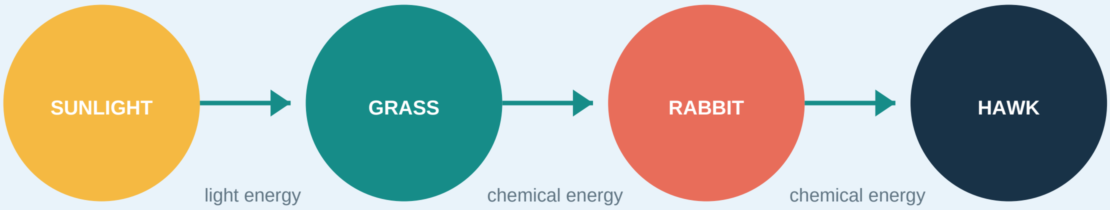


Think: Where would soil bacteria fit in this sequence? What abiotic factor might limit their growth?

Energy Moves, Matter Cycles

Energy enters most ecosystems as sunlight. Matter is reused as atoms move among organisms, soil, water, and air.

A simplified food chain



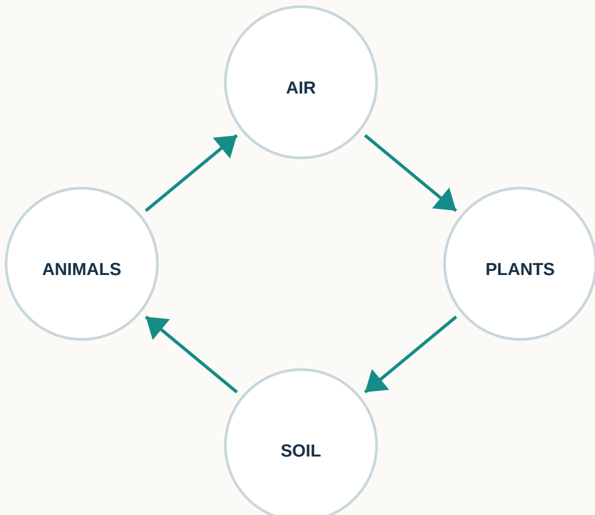
At each transfer, organisms use much of the available energy for movement, growth, repair, and maintaining body functions. Some energy leaves the system as heat.

Producers and consumers

Producers, such as plants and algae, capture light energy and store it in sugars. **Consumers** obtain energy by eating producers or other consumers. **Decomposers** break down wastes and remains, returning nutrients to the environment.

Reading the arrows
An arrow points from the energy source toward the organism that receives that energy. It does not show which organism is chasing another.

A matter cycle



Carbon atoms circulate through photosynthesis, feeding, respiration, waste, and decomposition.

INVESTIGATION 1.1

Light and Seedling Growth

Use evidence to evaluate how one abiotic factor can influence a living population.

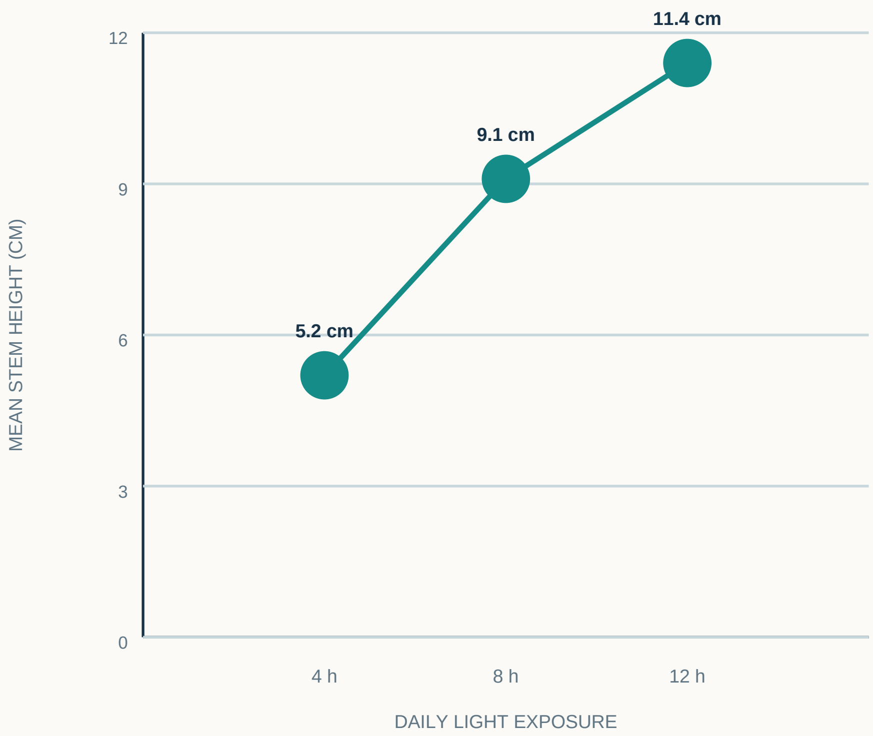
METHOD

1. Plant equal numbers of radish seeds in three identical trays.
2. Keep water, soil, and temperature constant.
3. Expose trays to 4, 8, or 12 hours of light daily.
4. After 14 days, measure mean stem height.

Recorded results

Daily light	Mean height
4 hours	5.2 cm
8 hours	9.1 cm
12 hours	11.4 cm

Visualizing the data



Analyze

1. Identify the independent and dependent variables.

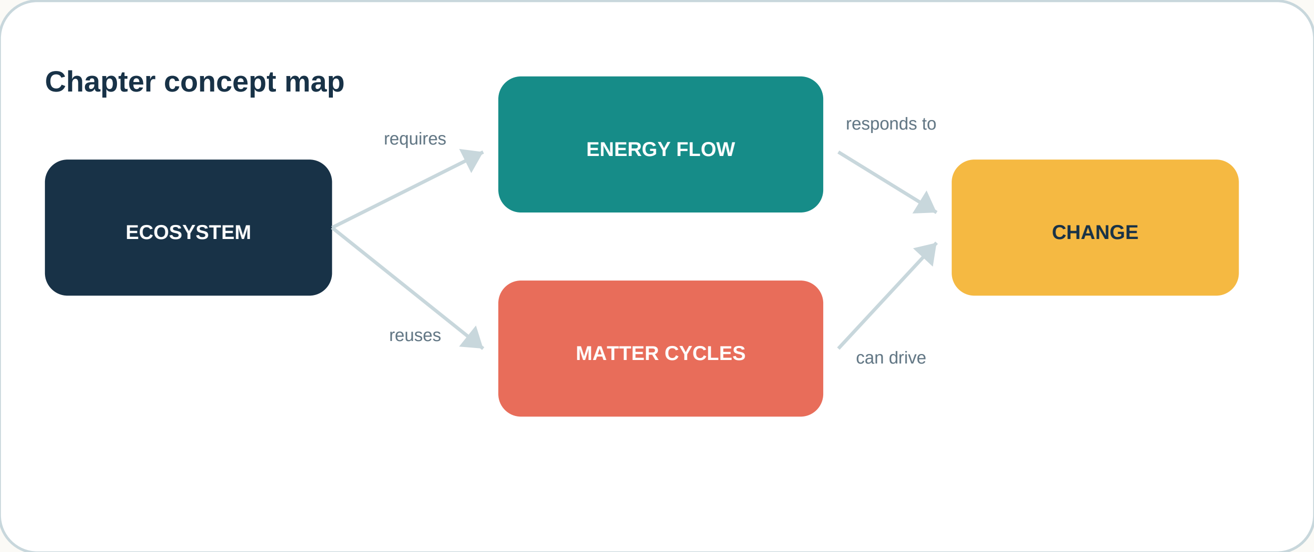
2. Describe the pattern shown in the graph.

3. Why must water, soil, and temperature remain constant?

4. Predict the result for 10 hours of daily light and justify your estimate.

Connect the Concepts

Use the chapter model to explain how organisms, energy, and matter form an interacting system.



Check your understanding

1. Distinguish between a population and a community using a pond example.
2. Classify sunlight, algae, dissolved oxygen, and fish as biotic or abiotic.
3. Explain why energy flows through an ecosystem while matter cycles.
4. Draw a food chain with a producer and two consumers. Add arrows in the correct direction.
5. A drought lowers plant growth. Predict one direct and one indirect effect on consumers.

SYSTEMS CHALLENGE

A new road divides a forest habitat. Fewer animals can travel between feeding and nesting areas, and rainwater carries sediment into a nearby stream.

Build an explanation

Use at least four terms from the word bank to describe how this change could affect the ecosystem.

population | community | abiotic | consumer | energy | habitat